



Optimizing staff scheduling in community emergency medical services using a self-learning genetic algorithm

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Received: 19 February 2025 / Revised: 10 March 2026 / Accepted: 10 April 2026 /
Published online: 12 May 2026

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2026

Abstract

Community healthcare has become a considerable portion of a country's economy due to its significant impact. One of the most important fields in community healthcare is the delivery of emergency medical services (EMS), which are provided to patients outside the hospital before their transfer to the nearest medical center. There is an increasing need to deliver the most effective care to individuals using limited resources. In this context, optimization methods have been successfully applied to manage various problems in healthcare environments. However, there are a limited number of research papers that consider community EMS as an optimization problem. To address this gap, this paper considers the staff scheduling problem in the Community EMS (SSP-CEMS) field, which is modeled as the vehicle routing problem with time windows (VRP-TW). Therefore, a self-learning genetic algorithm (SLGA) is proposed based on reinforcement learning. This strategy is used to automatically update the GA parameters to detect the next promising search direction. Then, computational experiments are performed using Solomon's benchmark instances. The obtained results demonstrate the competitive performance of the proposed SLGA relative to the clas-

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sical GA across standard VRP-TW benchmarks, supporting its potential applicability to community EMS scheduling contexts.

Keywords Emergency medical services · Community EMS · Optimization · Mathematical Modeling · Self-learning optimization

1 Introduction

Community Emergency Medical Services (CEMS) are considered an extension of traditional Emergency Medical Services (EMS). The latter plays a vital role in saving people's lives and reducing the rates of mortality and morbidity. EMS is a key component of community healthcare, as it is often the first point of contact with the healthcare system for many patients in need of urgent treatment. However, CEMS focuses more on preventive and community-based approaches in addition to emergency response. Therefore, CEMS may involve a wider range of personnel, including community health workers, social workers, nurses, and volunteers. Hence, the impact of staff scheduling in CEMS is significant and plays a crucial role in the effectiveness and efficiency of this healthcare delivery model.

Staff Scheduling Problems in CEMS (SSP-CEMS) are crucial for various aspects of community health. On the one hand, efficient scheduling ensures the optimal allocation of EMS personnel, including paramedics and community health workers, enhancing the community's ability to respond promptly to emergencies. In emergency situations, rapid response times are critical, and strategic staff scheduling ensures that EMS teams are readily available via an emergency telephone number, which connects them to a dispatching center. This proactive approach has the potential to save lives and minimize the impact of emergencies on community members. On the other hand, staff scheduling strategies contribute to supporting rural or underserved communities, where access to healthcare services may be limited. They provide regular access to healthcare providers through CEMS.

Furthermore, effective staff scheduling can optimize resource allocation, potentially leading to long-term cost savings. By addressing healthcare needs at the community level, CEMS can be considered a crucial part of home healthcare, which contributes to reducing overall healthcare costs.

Motivated by these issues, this work attempts to solve the staff scheduling problem in CEMS.

The considered problem is defined as a set of caregivers visiting patients at their locations to perform medical tasks while considering several constraints, especially their visiting time preferences. This problem is considered an extension of the well-known vehicle routing problem with time windows (VRP-TW), which aims to minimize the overall service time required to perform all medical tasks by caregivers at different locations. This problem falls into the category of NP-hard problems Cordeau et al. (2002).

In this context, operations research techniques have been successfully applied to manage various problems in healthcare environments. However, the number of papers that deal with SSP-CEMS is very limited. Consequently, in this paper, we present not

only the main studies that involve staff scheduling in different healthcare environments but also the well-known studies that focus on VRP-TW.

Moreover, this paper proposes a self-learning genetic algorithm based on a reinforcement learning strategy. Reinforcement learning (RL) is a knowledge-driven paradigm that can directly guide one or more agents to select appropriate actions based on experience gained from interactions with the environment.

The main contributions of this paper are summarized as follows:

1. Proposing a new mathematical formulation for the staff scheduling problem in CEMS (SSP-CEMS), which can be considered a variant of VRP with time windows (VRP-TW).
2. Proposing a new self-learning genetic algorithm, called SLGA, to solve the SSP-CEMS.
3. Demonstrating the superior performance of our SLGA on different commonly used benchmark problems from the VRP-TW literature.

The remainder of this paper is structured as follows. Section 2 presents an overview of the literature on VRP-TW and staff scheduling problems in different healthcare environments, with particular emphasis on SSP-CEMS systems. Next, the problem definition is presented in Section 3, followed by the proposed solution approach in Section 4. In Section 5, the proposed method is empirically evaluated and compared with other methods. Finally, Section 6 concludes the paper and outlines future research opportunities.

2 Related Works

Healthcare systems have played a crucial role in the automated personnel scheduling research literature, mainly due to the importance of staff scheduling in home healthcare (HHC), which has been widely studied. In this context, the problem is generally defined as a set of caregivers providing a wide range of medical and paramedical services to patients in their homes, such as nursing care, drug delivery, and medical assistance.

The HHC staff scheduling problem was investigated by Mutingi and Mbohwa (2014), who proposed a clustering particle swarm optimization (CPSO) method aiming to optimize the balance of workload among staff members. Similarly, Restrepo et al. (2020) studied scheduling in home healthcare by proposing a two-stage stochastic programming model. In this model, first-stage decisions correspond to staffing and scheduling caregivers in geographic districts, whereas second-stage decisions concern the temporary reallocation of caregivers to neighboring districts.

Another study by Shiri et al. (2021) introduced a novel hybrid three-phase methodology for routing and scheduling in home healthcare. The proposed framework includes three main stages: (1) ranking candidate locations for opening healthcare facilities based on qualitative criteria, (2) addressing uncertainty in service times, and (3) incorporating quantitative decision factors. Furthermore, Demirbilek et al. (2019) examined the home healthcare nurse scheduling problem in a dynamic environment where patients arrive over time and appointment decisions must be made immediately

upon their arrival. The objective of this work is to maximize the average number of daily visits performed by a single nurse.

The role of technology in improving healthcare accessibility was highlighted by Euchi et al. (2020), who discussed the importance of telemedicine and digital solutions for serving populations with limited access to healthcare services. The authors proposed an AI-based method for scheduling and routing home healthcare services and demonstrated its efficiency and cost-effectiveness. Their approach is based on generating several scheduling scenarios that include both randomly generated and real patient requests.

More recently, Xiang et al. (2023) proposed a hybrid elitist non-dominated sorting genetic algorithm (NSGA-II) enhanced with a local search procedure to address daily routing and scheduling problems in HHC. The proposed approach simultaneously minimizes operational costs and maximizes participant preference satisfaction. In a related work, Belhor et al. (2023) introduced a multi-objective evolutionary approach based on K-means clustering. Two hybrid algorithms were proposed, combining NSGA-II and the Strength Pareto Evolutionary Algorithm (SPEA2) with clustering techniques. In this approach, the population is divided into clusters, and the sub-solutions are later combined to construct the final Pareto front. For a broader overview of routing and scheduling problems in home healthcare environments, readers may refer to the survey and bibliometric analysis presented in Di Mascolo et al. (2021).

The management of healthcare resources also became particularly critical during the COVID-19 pandemic. In this context, Aydemir-Karadag (2022) proposed a bi-objective mixed-integer nonlinear programming model for healthcare waste management. The model integrates location decisions with waste collection scheduling through a two-step optimization framework.

Emergency medical services represent another essential component of healthcare systems, playing a vital role in stabilizing and transporting critically injured patients before hospital arrival. EMS systems aim to respond rapidly to emergency calls, provide essential first aid, and transport patients to appropriate hospitals. One of the most widely studied optimization problems in EMS is the ambulance location and routing problem. For instance, Zhu et al. (2023) studied the integrated problem of ambulance dispatching and operating room scheduling during mass casualty incidents. Similarly, Khoshghebari et al. (2023) proposed new heuristic approaches combined with a two-stage stochastic integer programming model to address ambulance location and routing under multiple sources of uncertainty.

Comprehensive surveys on ambulance location and routing problems are presented in Tassone and Choudhury (2020). Additional reviews of EMS operations research applications can be found in Aboueljainane et al. (2013) and Aringhieri et al. (2017). Furthermore, Yu et al. (2023) investigated macro-scheduling problems for mixed helicopter fleets in natural disaster scenarios related to EMS. The authors proposed a hierarchical optimization approach that decomposes the scheduling process into mission assignment and route planning stages.

Despite the large body of research on staff scheduling problems, relatively few studies specifically address scheduling issues within EMS systems. For example, Addis et al. (2012) studied the EMS system in Milan, where personnel must be assigned to teams and shifts while considering forecasted demand. Similarly, Rajagopalan et al.

(2011) proposed a two-stage solution method that integrates ambulance location decisions with the scheduling of ambulance crew working hours to maximize expected coverage. Another relevant contribution is presented in Vermuyten et al. (2018), who proposed an integer programming formulation together with two heuristic approaches for solving staff scheduling problems in an EMS system operating in Portugal.

To the best of our knowledge, very limited research has addressed staff scheduling in Community Emergency Medical Services (CEMS) from an optimization perspective. Most existing studies focus either on home healthcare scheduling or on ambulance routing and dispatching problems. However, the integration of staff scheduling decisions within CEMS environments remains largely unexplored. Motivated by this research gap, this paper formulates the CEMS staff scheduling problem as an optimization problem modeled as an extension of the vehicle routing problem with time windows (VRP-TW).

Due to the complexity of the VRP-TW, many studies have proposed approximate solution approaches. Readers may refer to the well-known surveys by Bräysy et al. (2004), Bräysy and Gendreau (2005), and Dixit et al. (2018) for comprehensive reviews of solution techniques for this class of problems.

Among the most widely used approximate methods are evolutionary algorithms (EAs). These algorithms are particularly suitable for solving complex combinatorial optimization problems such as VRP due to their ability to explore large search spaces and handle multiple decision variables Tan et al. (2023). For instance, Maroof et al. (2024) proposed a hybrid genetic algorithm combined with the Solomon insertion heuristic for solving the VRP-TW. In another study, Joshi and Thakur (2023) introduced two nature-inspired routing mechanisms designed to optimize throughput and latency in on-chip networks, demonstrating significant improvements in network performance.

Several works have also explored hybrid evolutionary strategies. For example, Zhang et al. (2022b) investigated a multi-objective evolutionary algorithm that integrates global search mechanisms with problem-specific local search strategies to achieve faster convergence during different evolutionary stages.

The performance of evolutionary algorithms is highly sensitive to the configuration of their control parameters. Consequently, recent research has increasingly focused on adaptive mechanisms that dynamically adjust operator selection and algorithm parameters during the search process. These adaptive strategies allow algorithms to modify their behavior according to the environment in which they operate. For example, Zhang et al. (2022a) investigated a multi-objective evolutionary algorithm based on adaptive local search chains. Similarly, Sabar et al. (2019) proposed an evolutionary algorithm capable of evolving configurations that include parameter values, operator types, and operator sequences. These configurations are encoded within dynamic VRP solutions, enabling the algorithm to adapt its search strategy during the optimization process. Another adaptive hybrid approach combining a genetic algorithm with an artificial bee colony algorithm was proposed by Liu (2013), where the crossover and mutation ratios are dynamically adjusted and the fitness function incorporates an adaptive penalty coefficient.

More recently, research has explored self-learning mechanisms to improve algorithm performance over time. These approaches rely on learning from historical data to guide the search process. In particular, reinforcement learning (RL) has emerged as a promising framework capable of adapting exploration strategies based on reward feedback while simultaneously learning optimal decision policies. For example, Nazari et al. (2018) proposed a reinforcement learning approach for solving the VRP by training a policy model capable of generating near-optimal solutions for a wide range of problem instances. A comprehensive survey on reinforcement learning approaches for VRP can be found in Raza et al. (2022).

Furthermore, several studies have investigated the integration of reinforcement learning within metaheuristic algorithms. For instance, Chen et al. (2020) proposed a hybrid method combining genetic algorithms with reinforcement learning to solve the flexible job shop scheduling problem, where RL is used to dynamically adjust GA parameters. Similarly, Sun et al. (2022) introduced a hybrid approach combining tabu search with reinforcement learning to solve the minimum load coloring problem. Another study by Masadeh et al. (2023) explored several reinforcement learning algorithms based on the selector–actor–critic framework.

For a broader overview of self-learning strategies in optimization algorithms, readers may refer to the survey presented in Macías-Escrivá et al. (2013).

3 Problem description and mathematical formulation

The Staff Scheduling Problem in Community Emergency Medical Services (SSP-CEMS) integrates staff scheduling decisions with emergency healthcare operations. Its main objective is to efficiently allocate medical staff and vehicles to provide healthcare services or deliver medical supplies to multiple healthcare facilities within a city while minimizing the total travel distance or travel time.

The main scenario of the considered SSP-CEMS is illustrated in Figure 1.

As shown in Figure 1, the staff scheduling process in Community Emergency Medical Services involves several sequential stages described as follows:

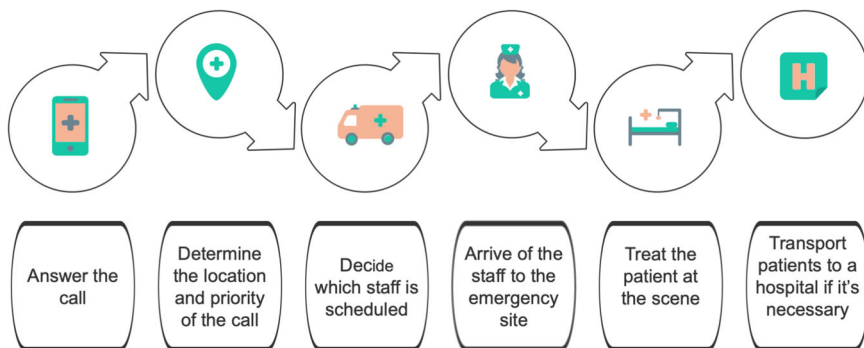


Fig. 1 Staff scheduling in the Community Emergency Medical Services scenario description

1. **Answer the call:** The process begins by receiving and responding to emergency calls from patients requiring medical assistance.
2. **Determine the location and priority of the call:** Once the call is received, the emergency dispatcher identifies the patient's location and determines the priority level of the emergency based on the severity of the case.
3. **Determine the staff to be scheduled:** After evaluating the emergency's location and priority, the dispatcher selects the most appropriate medical staff to respond. This decision is based on several factors, such as proximity, staff expertise, and availability.
4. **Arrival of the staff at the emergency site:** Once the staff members are assigned, they travel to the emergency location to ensure a timely response and provide the required medical assistance.
5. **Treat the patient at the scene:** Upon arrival, the medical staff evaluate the patient's condition and provide immediate treatment on-site.
6. **Transport patients to a hospital if necessary:** If the patient requires further treatment, the medical staff arrange transportation to a hospital to ensure continuity of care.

The problem consists of determining optimal routes and schedules for vehicles departing from a central medical depot to visit several healthcare facilities. Each facility i has a demand q_i and a service time window $[s_i, e_i]$. Vehicles have a limited capacity Q , and traveling between locations incurs travel distances or travel times d_{ij} .

The scheduling process must ensure that each facility is visited exactly once while respecting time window and vehicle capacity constraints. The optimization objective is to minimize the total travel distance while ensuring efficient service delivery to all healthcare facilities.

This problem can be modeled as a variant of the well-known Vehicle Routing Problem with Time Windows (VRP-TW), which aims to minimize the total service time required for caregivers to perform medical tasks at different locations.

To mathematically formulate the SSP-CEMS, the following sets, parameters, and decision variables are defined.

Sets

$V = \{0, 1, 2, \dots, n\}$: Set of vertices representing the depot and healthcare facilities.

$A = \{(i, j) \mid i, j \in V, i \neq j\}$: Set of arcs representing possible routes between vertices.

Parameters

d_{ij} Travel distance (or time) between vertices i and j .

q_i Number of medical staff required at facility i .

s_i, e_i Start and end of the service time window for facility i .

Q Capacity of each vehicle (maximum number of staff that can be transported).

M A sufficiently large positive constant.

t_{ij} Travel time from vertex i to vertex j .

p_i Service time at facility i .

Decision Variables

x_{ij} Binary variable equal to 1 if a vehicle travels from vertex i to vertex j , and 0 otherwise.

u_i Arrival time of the vehicle at facility i .

The mathematical model for the SSP-CEMS is formulated as follows.

Objective Function

$$\min \sum_{(i,j) \in A} d_{ij} x_{ij} \quad (1)$$

Subject to

$$\sum_{j \in V, j \neq i} x_{ij} = 1 \quad \forall i \in V \setminus \{0\} \quad (2)$$

$$\sum_{i \in V, i \neq j} x_{ij} = 1 \quad \forall j \in V \setminus \{0\} \quad (3)$$

$$\sum_{j \in V, j \neq i} x_{ij} - \sum_{j \in V, j \neq i} x_{ji} = 0 \quad \forall i \in V \quad (4)$$

$$s_i \leq u_i \leq e_i \quad \forall i \in V \quad (5)$$

$$\sum_{i \in V} q_i x_{ij} \leq Q \quad \forall j \in V \setminus \{0\} \quad (6)$$

$$u_j \geq u_i + t_{ij} - M(1 - x_{ij}) \quad \forall (i, j) \in A, i \neq 0, j \neq 0 \quad (7)$$

$$u_j \geq s_j \quad \forall j \in V \setminus \{0\} \quad (8)$$

Variable domains

$$x_{ij} \in \{0, 1\} \quad \forall (i, j) \in A \quad (9)$$

$$u_i \geq 0 \quad \forall i \in V \quad (10)$$

Equation (1) represents the objective function that minimizes the total travel distance between healthcare facilities. Equations (2) and (3) ensure that each facility is visited exactly once. Specifically, Equation (2) guarantees that exactly one arc leaves each facility, while Equation (3) ensures that exactly one arc enters each facility. Equation (4) represents the flow conservation constraint, ensuring route continuity by maintaining the balance between incoming and outgoing arcs at each vertex. Equation (5) imposes the time window constraints, ensuring that the arrival time at each facility occurs within its predefined service time interval. Equation (6) ensures that vehicle capacity limits are respected by restricting the total demand served by each vehicle. Equation (7) represents the time propagation constraint, which guarantees the temporal feasibility of the routes by linking the arrival times of consecutive facilities.

Finally, Equation (8) ensures that the arrival time respects the earliest possible service time at each facility.

4 Solution framework

Self-Learning Genetic Algorithm (SLGA) is a variant of the traditional genetic algorithm that incorporates self-adaptation mechanisms to dynamically adjust its parameters during the optimization process. Unlike genetic algorithms (GAs), which rely on fixed parameters set by the user, SLGA autonomously tunes its parameters based on the performance of the evolving population. This self-adjustment capability allows SLGA to adapt to changes in the problem landscape and improve its effectiveness in finding optimal solutions.

4.1 The proposed genetic algorithm

The basic concept of a GA involves simulating the process of evolution through the use of genetic operators, including mutation, crossover, and selection. The algorithm begins with a population of potential solutions encoded as chromosomes, each representing a candidate solution to the problem.

Algorithm 1 details the main steps of a basic GA.

```

Input: Initial population, termination condition
Output: Best solution
Create a random population;
Compute the fitness of each individual in the population;
while termination condition not met do
    Determine optimal parameter settings for each individual using reinforcement learning;
    Select two parents from the population based on their fitness;
    Perform genetic crossover to produce offspring with optimal parameters;
    Mutate offspring chromosomes based on optimal parameters;
    Insert mutated offspring into the new population;
    Evolve parameter settings for each individual in the population;
    Replace the population;
end
return Best solution;
  
```

Algorithm 1: Genetic Algorithm

A GA explores the search space by iteratively evolving the population through selection, crossover, and mutation. Over time, better solutions tend to emerge through the process of natural selection, leading to an improved population of individuals Verma and Kumar (2014). In a GA population, there are n chromosomes, each of which has a fitness value. Offspring are generated from parents selected based on their fitness. Therefore, individuals with better fitness values have a higher chance of survival. An initial population is typically generated randomly or using heuristic methods to improve solution quality.

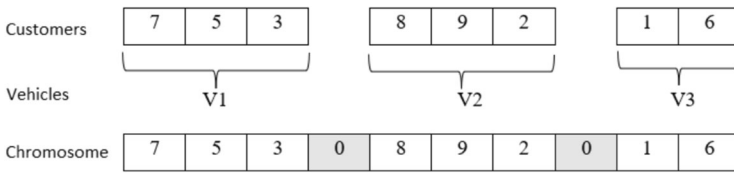


Fig. 2 Chromosome encoding of customer visits.

In our solution approach, the GA iteratively applies three operators after generating a random population sample:

- **Tournament selection:** equivalent to the survival of the fittest.
- **Ordered crossover:** which symbolizes cross-species mating.
- **Inversion mutation:** which introduces primarily random changes.

In what follows, we outline the solution encoding, the initial population, and the GA operators employed.

4.1.1 Solution encoding

A chromosome in this GA consists of n genes, each representing a node. An individual represents a set of homogeneous vehicles responsible for visiting these nodes. In addition, the depot is included in the chromosome and represents the start and end of each vehicle route. The sequence between two “0” values corresponds to the sequence of nodes visited by a vehicle (Fig. 2.).

4.1.2 Initial population

In each generation, the population is a subset of possible solutions (i.e., a set of chromosomes). Two common practices are used to initialize a GA population: random initialization and heuristic initialization. In our case, the population is generated randomly. A vector is used to represent each solution. Thus, we first generate a random population of N solutions, and then we define the fitness function used to evaluate them.

4.1.3 Fitness function

A chromosome represents each solution. These solutions must be evaluated to identify the best candidates. Accordingly, each solution is assigned a fitness score to determine the likelihood of being selected for reproduction. Fitter solutions have a higher chance of survival, whereas weaker ones are eliminated. This mechanism supports convergence. In the developed GA, we use the objective function defined in the mathematical formulation, which aims to minimize the total cost.

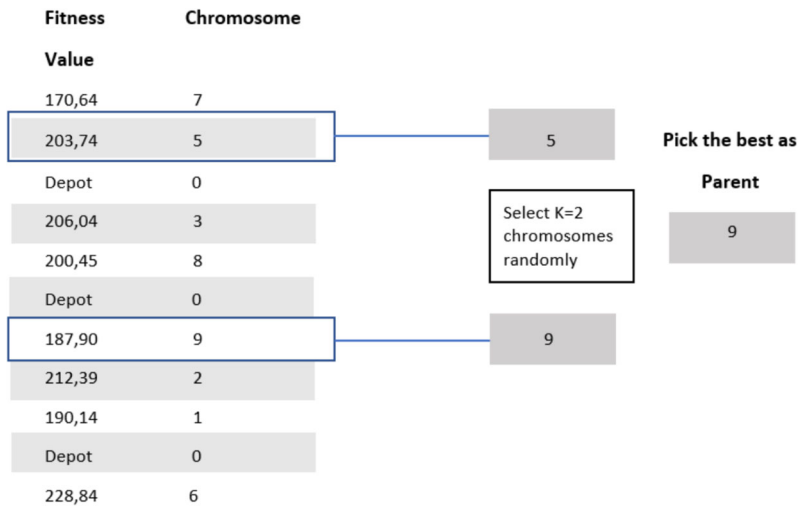


Fig. 3 Tournament selection.

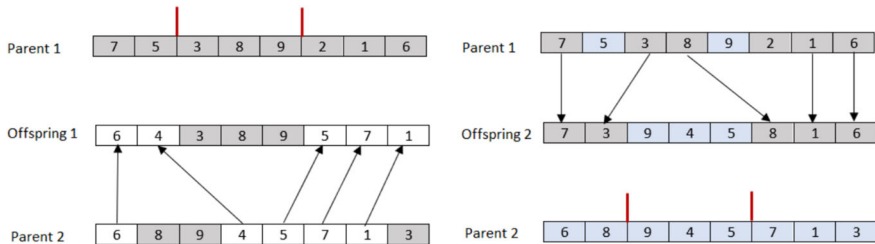


Fig. 4 Ordered crossover.

4.1.4 Selection technique

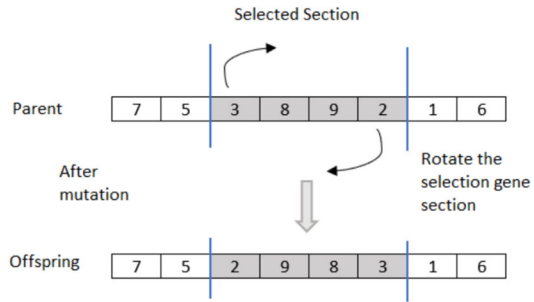
We adopt tournament selection. This method consists of selecting $K = 2$ individuals randomly from the population and choosing the best among them as a parent, as shown in Figure 3.

4.1.5 Ordered crossover

In this work, we implement ordered crossover (OX). Two parent chromosomes are selected using tournament selection. Then, two random crossover points are chosen, dividing the chromosome into three parts (left, middle, and right). The two-point OX operates as follows: child1 inherits its left and right parts from parent1, while its middle part is completed according to the order of genes in parent2.

4.1.6 Inversion mutation

To generate the new generation, the two offspring produced by each crossover operation are submitted to the mutation operator. Mutation changes one or more positions

Fig. 5 Inversion mutation.

within a chromosome to maintain diversity. In this study, we use inversion mutation, which randomly selects two positions in a chromosome and inverts the substring between them (Fig. 5).

4.1.7 Elitism method

We employ an elitism insertion strategy, which copies the best chromosome from the previous population into the new one. To maintain a constant population size from one generation to the next, this solution is complemented by the offspring produced through crossover and mutation.

4.2 Self-Learning Genetic Algorithm

SLGA employs adaptive parameters at both the population and individual levels to automatically adjust parameters such as the population size and mutation frequency during the search process. The self-learning nature of mutation strength has been investigated in previous studies Hoffmeister and Bäck (1990); Hinterding (1995).

Figure 6 illustrates how the parameters of a genetic algorithm can be encoded into each individual's chromosome.

(a) Each individual chromosome consists of two parts separated by a double bar: route genes representing the sequence of visited nodes (blue cells) with depot separators (grey cells), and four self-adaptive strategy parameter genes (orange cells)—mutation rate μ , mutation step size σ , population size N , and crossover rate P_c , which are co-evolved alongside the route genes throughout the evolutionary process.

(b) Self-adaptation feedback mechanism. Fitness values computed during evaluation are used to derive a reward signal r for the reinforcement learning agent. At each generation, the RL agent observes the current state s (characterized by population diversity and stagnation detection) and selects a parameter configuration (μ, σ, N, P_c) according to the Q-learning update rule $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$, with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.9$. The selected parameters are applied to the genetic operators (tournament selection, ordered crossover, and inversion mutation). The resulting offspring form the new population, with the best individual preserved by elitism (dashed arrow). This cycle repeats

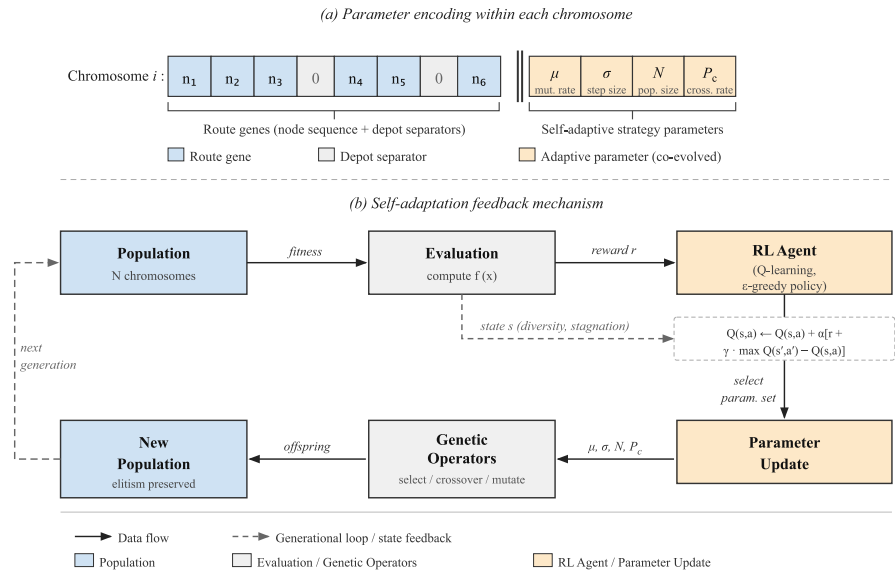


Fig. 6 Self-Adaptive Genetic Algorithm (SAGA) parameter encoding framework.

each generation, allowing SLGA to autonomously adapt its search behavior based on runtime feedback.

- The individual exhibiting the most favorable adaptation to its environment transmits its genetic parameters to the next generation.
- Parameter values are determined entirely by the evolutionary process.
- The development of an autonomous individual is achieved through parameter self-adaptation.

Reinforcement learning (RL) is a subfield of machine learning that relies on rewarding desirable actions and penalizing undesirable ones. Agents act in ways that either increase or decrease the reward. RL performs learning through a trial-and-error process. Upon each interaction between an agent and the environment, the environment provides a new state, prompting the agent to transition accordingly. Undesirable behaviors incur a negative value as a penalty, while positive values reward favorable actions. These scalar values provide feedback, informing the agent about the success or failure of its actions.

The purpose of this strategy is either to enhance metaheuristics with machine learning or, conversely, to improve the execution of machine learning approaches using metaheuristics. To solve SSP-CEMS, where model inputs appear either in the objective function or in the set of constraints, our learn-heuristic framework follows this principle.

Figure 7 illustrates the core framework of the proposed approach. To build machine learning prediction models, historical data and related inputs (e.g., customer demands) are first used. Then, when the solution structure changes, these models are applied iteratively throughout the learn-heuristic construction process to generate updated

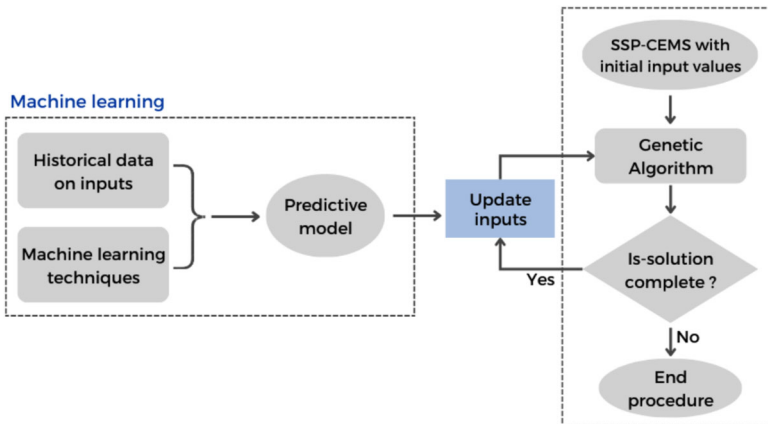


Fig. 7 SLGA framework.

estimations of the problem inputs. A complete solution is produced after the construction process is completed, noting that there is a risk of obtaining suboptimal results without the learning mechanism.

The following pseudo-code provides a more detailed explanation of the main learn-heuristic framework. At each iteration, the algorithm assesses the state of the partially constructed solution, updates the problem inputs according to this state using the predictive model, and then, depending on the updated inputs, proceeds to the next step of solution construction by applying heuristic logic.

5 Numerical results

This section presents the experimental evaluation of the proposed Self-Learning Genetic Algorithm (SLGA) and compares its performance with the classical Genetic Algorithm (GA). The experimental study is organized as follows. First, the benchmark instances and experimental setup are described. Next, parameter tuning is performed using the Taguchi method. Then, comparative results are presented. Finally, robustness, computational efficiency, and statistical significance analyses are provided.

5.1 Benchmark instances

The experimental evaluation is conducted using benchmark instances derived from the well-known Solomon datasets, which are widely used in the literature for evaluating vehicle routing algorithms. The datasets are available at:

<http://neo.lcc.uma.es/vrp/vehicle-routing-problem/>

Solomon's benchmark instances represent different spatial distributions of customers and are commonly classified into three categories:

- C1 and C2: clustered customer locations
- R1 and R2: randomly distributed customer locations

- RC1 and RC2: mixed distributions combining clustered and random locations.

In this study, a representative subset of instances with up to 100 customers is selected to evaluate the effectiveness of the proposed algorithm.

Experiments were conducted on a PC equipped with an Intel Pentium IV processor (1.6 GHz) and 512 MB of RAM.

Because metaheuristic algorithms are stochastic in nature, each instance was executed in **30 independent runs**. The best, mean, and standard deviation of the obtained solutions were recorded to evaluate solution quality and algorithmic stability.

The reinforcement learning component embedded in SLGA updates the problem inputs during the solution construction process based on the reward signals obtained from intermediate solutions.

To ensure the reliability of the experimental results, all algorithms were implemented in the same programming environment and executed under identical computational conditions. Each instance was solved 30 independent times in order to account for the stochastic nature of metaheuristic algorithms. The reported statistics therefore provide a fair and reproducible comparison between the proposed SLGA and the classical GA.

5.2 Parameter tuning

In this study, Taguchi tuning is applied to determine suitable initial parameter settings for the baseline GA. In contrast, the proposed SLGA dynamically adapts its parameters during the evolutionary process through the self-learning mechanism described in Section 4.

Parameter settings strongly influence the performance of metaheuristic algorithms. In this study, parameter tuning was performed using the Taguchi Design of Experiments (TDOE) method.

The Taguchi method uses orthogonal arrays to evaluate the effect of parameters while minimizing the number of required experiments. Mutation parameters are self-adaptive in SLGA and therefore are not included in the Taguchi tuning procedure. An *L9* orthogonal array was employed to analyze three parameters at three levels:

- Number of iterations
- Population size
- Crossover rate

Table 1 presents the considered parameter levels.

Table 1 Considered levels of parameters

	Parameter	Level 1	Level 2	Level 3
GA/SLGA	Number of iterations	100	200	300
	Population size	50	100	150
	Crossover rate	0.7	0.8	0.9

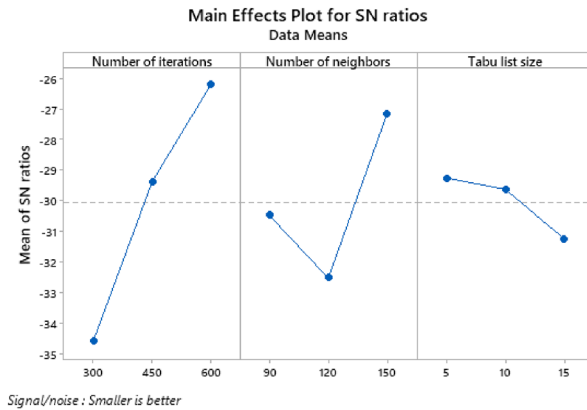


Fig. 8 Mean S/N ratio for GA.

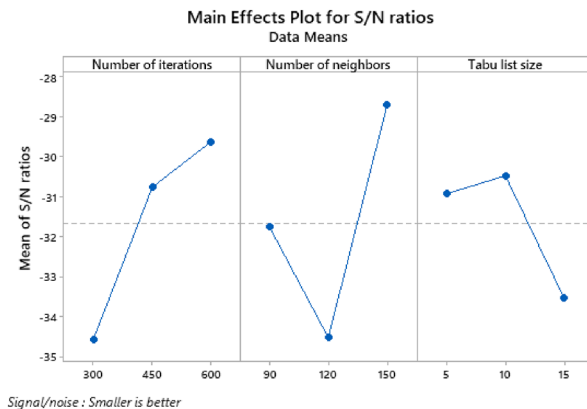


Fig. 9 Mean S/N ratio for SLGA.

The Signal-to-Noise (S/N) ratio was used to determine the most robust parameter configuration. Higher S/N values indicate better parameter settings. The results are illustrated in Figures 8 and 9.

5.3 Comparative study

Table 2 reports the comparative performance of GA and SLGA regarding the Best Known Solutions (BKS). The reported values correspond to the **best solutions obtained over 30 independent runs**.

The results prove that SLGA consistently produces solutions closer to the best-known solutions. The average gap is reduced from 1.33% with GA to only 0.25% with SLGA.

Table 2 Comparative results: GA vs SLGA

Instance	BKS		GA			SLGA		
	#veh	Route	#veh	Route	GAP(%)	#veh	Route	GAP(%)
r103	13	1292.68	13	1293.40	0.06	13	1292.90	0.02
r112	9	982.14	9	1004.10	2.24	9	984.50	0.24
r201	4	1253.23	4	1256.80	0.28	4	1254.10	0.07
r202	3	1202.53	3	1217.90	1.28	3	1206.20	0.30
r204	2	856.36	2	868.70	1.44	2	858.10	0.20
r206	3	906.14	3	915.20	1.00	3	908.40	0.25
r207	2	890.10	2	902.50	1.39	2	893.20	0.34
r208	2	726.82	2	740.10	1.82	2	729.10	0.31
r209	3	921.66	3	927.30	0.61	3	923.10	0.15
r210	3	939.37	3	968.80	3.13	3	945.20	0.62

Table 3 Percentage improvement of SLGA over GA

Instance	Improvement (%)
r103	0.04
r112	1.95
r201	0.21
r202	0.96
r204	1.22
r206	0.74
r207	1.03
r208	1.49
r209	0.45
r210	2.44
Average improvement	1.05

5.4 Improvement analysis

To quantify the advantage of the proposed SLGA over the classical GA, the percentage improvement in route length was computed for each instance using the following formula:

$$Improvement(\%) = \frac{Route_{GA} - Route_{SLGA}}{Route_{GA}} \times 100$$

Table 3 reports the improvement obtained by SLGA relative to GA.

The results indicate that the proposed SLGA consistently improves the solutions obtained by the classical GA. The improvement ranges from 0.04% for instance r103 to 2.44% for instance r210, with an average improvement of approximately 1.05% across all tested instances. These results demonstrate that the self-learning mechanism

effectively enhances the exploration and exploitation capabilities of the genetic search process.

These improvements are further validated by the statistical analysis presented in Section 5.9, confirming that the observed differences are statistically significant.

5.5 Robustness analysis

To evaluate algorithm stability, Table 4 reports the best, mean, and standard deviation obtained across 30 independent runs.

SLGA exhibits lower variability across most instances, indicating greater robustness.

5.6 CPU time analysis

Table 5 reports the average CPU time over 30 runs.

Table 4 Best, mean, and standard deviation over 30 runs

Instance	GA			SLGA		
	Best	Mean	Std	Best	Mean	Std
r103	1293.40	1300.61	9.28	1292.90	1295.01	2.32
r112	1004.10	1006.42	3.46	984.50	985.23	1.62
r201	1256.80	1260.61	5.73	1254.10	1255.18	1.51
r202	1217.90	1218.46	3.66	1206.20	1207.15	2.53
r204	868.70	869.75	2.40	858.10	858.93	2.18
r206	915.20	922.63	10.31	908.40	909.12	2.34
r207	902.50	915.07	11.95	893.20	898.86	2.86
r208	740.10	742.02	3.57	729.10	729.84	1.44
r209	927.30	929.07	5.49	923.10	923.67	1.73
r210	968.80	971.11	7.69	945.20	946.31	2.69

Table 5 CPU time (Mean $\pm$ Std)

Instance	GA	SLGA
r103	2.17 $\pm$ 0.14	8.36 $\pm$ 0.75
r112	3.93 $\pm$ 0.31	8.31 $\pm$ 0.39
r201	2.54 $\pm$ 0.13	9.30 $\pm$ 0.55
r202	3.50 $\pm$ 0.31	6.19 $\pm$ 0.62
r204	1.43 $\pm$ 0.07	10.49 $\pm$ 0.69
r206	2.85 $\pm$ 0.20	5.04 $\pm$ 0.35
r207	2.96 $\pm$ 0.18	7.82 $\pm$ 0.44
r208	1.87 $\pm$ 0.11	9.63 $\pm$ 0.51
r209	3.12 $\pm$ 0.22	6.98 $\pm$ 0.47
r210	3.41 $\pm$ 0.25	8.15 $\pm$ 0.58

Although SLGA requires more computational time, the improvement in solution quality justifies the additional computational cost.

5.7 Performance profile

To further analyze the performance of the algorithms, a Dolan–Moré performance profile was constructed based on the mean route length values (Fig. 10).

The performance profile indicates that SLGA dominates GA across the tested instances.

5.8 Convergence analysis

To illustrate the convergence behavior of the algorithms, Figure 11 shows the evolution of the best-so-far route length over iterations on a representative instance (r210). SLGA converges to a better solution and stabilizes earlier than GA, indicating a more effective balance between exploration and exploitation.

5.9 Statistical analysis

A comprehensive statistical analysis was conducted to evaluate whether the observed performance differences between GA and SLGA are statistically significant.

5.9.1 Global comparison

A paired Wilcoxon signed-rank test was applied to compare the performance of GA and SLGA across all instances, using the mean route lengths obtained from 30 independent

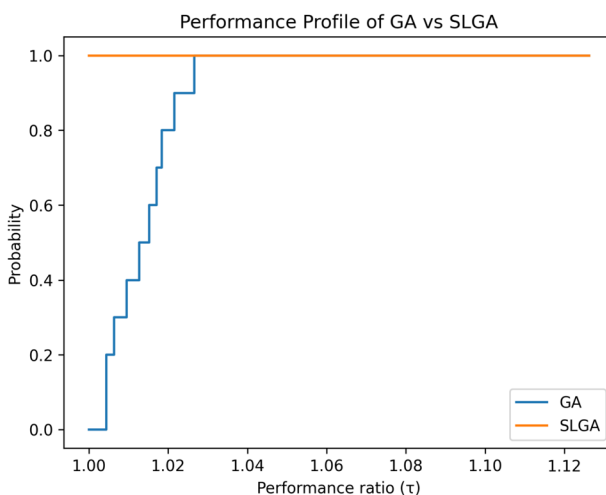


Fig. 10 Performance profile comparing GA and SLGA

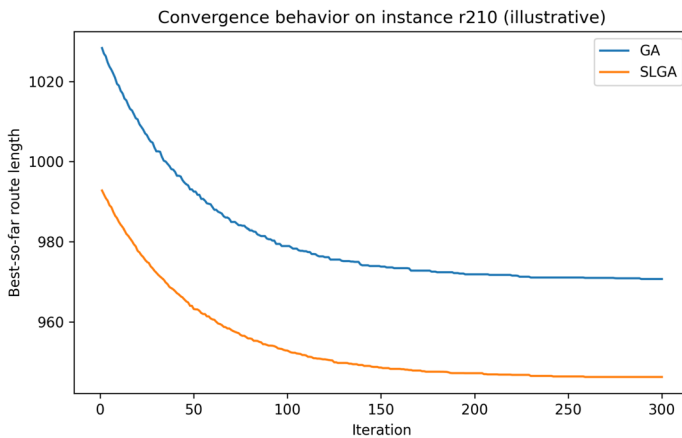


Fig. 11 Convergence behavior (best-so-far route length) of GA and SLGA on instance r210.

Table 6 Instance-wise Wilcoxon test p-values

Instance	p-value
r103	0.003
r112	0.001
r201	0.052
r202	0.008
r204	0.042
r206	0.002
r207	0.004
r208	0.001
r209	0.061
r210	0.001

runs. This non-parametric test is appropriate for pairwise comparisons and does not assume normality of the data.

The null hypothesis (H_0) assumes that there is no difference between the algorithms, while the alternative hypothesis (H_1) assumes that SLGA produces better solutions.

The test yielded a p-value of:

$$p = 0.00071 < 0.05$$

Therefore, the null hypothesis is rejected at the 95% confidence level, confirming that SLGA significantly outperforms GA globally.

5.9.2 Instance-wise analysis

To identify which specific instances show statistically significant improvements, pairwise Wilcoxon tests were performed for each instance using the 30 run results. Table 6 presents the p-values for each instance.

As shown in Table 6, SLGA achieves statistically significant improvements ($p < 0.05$) in 8 out of 10 instances. For instances r201 and r209 ($p = 0.052$ and $p = 0.061$, respectively), the improvements are consistent but do not reach statistical significance at the $\alpha = 0.05$ level. This may be attributed to the particular characteristics of these instances or slightly higher variability in the results.

These instance-wise results complement the global analysis and demonstrate that the superiority of SLGA is consistent across most test problems.

6 Conclusion and future research

Community Emergency Medical Services (CEMS) play a vital role in saving lives and reducing mortality and morbidity rates. Emergency Medical Services (EMS) are a key component of community healthcare, as they are often the first point of contact with the healthcare system for patients requiring urgent treatment. In this context, optimization methods and techniques have been successfully applied to address a wide range of problems in healthcare environments.

In this paper, we addressed the staff scheduling problem in the community EMS field by proposing a self-learning genetic algorithm. The computational results demonstrate the effectiveness of the proposed approach. Moreover, the comparative analysis validates the performance of our model, highlighting its potential to provide decision support for community EMS operations.

Funding The authors extend their appreciation to the Deputyship for Research & Innovation, Ministry of Education in Saudi Arabia for funding this research work through the project number RI-44-0008.

Data Availability The datasets generated and analyzed during the current study are available upon request.

Declarations

Conflicts of Interest The authors have no relevant financial or non-financial interests to disclose.

Ethical and informed consent for data used This study was conducted following the ethical standards outlined in the relevant ethical guidelines. Informed consent was obtained from all authors included in the study. The purpose and procedures of the research were explained, and participants provided written consent.

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